

CDN Multi-Metric Server Selection: Key Findings

□ CORE PROBLEM

- RTT explains only 2.6% of throughput variance ($R^2 = 0.026$)
- Correlation $r = -0.161$ (weak but statistically significant, $p < 0.001$)
- 97.4% of throughput variance remains unexplained by RTT alone

□ MODEL PERFORMANCE

- Linear Model (A): $R^2 = 5.7\%$ on test set
- Ridge Regression (B): $R^2 = 11.7\%$ on test set
- Neural Network (C): $R^2 = 20.5\%$ on test set (best)
- All models predict in log-space for stability

□ SERVER SELECTION RESULTS

- RTT-Only: 142.9 Mbps median (BEST)
- Model A: 123.3 Mbps (-13.7% vs RTT)
- Model B: 122.3 Mbps (-14.4% vs RTT)
- Model C: 140.3 Mbps (-1.8% vs RTT)
- Conclusion: Limited features (RTT + loss) insufficient for ML improvement

□ 5G MOBILE NETWORKS

- RSRP explains 22.3% variance (9x better than RTT!)
- Multi-metric (RSRP+RSRQ+SINR): 24.3% R^2
- Mobile shows more promise for multi-metric selection

□ ADDITIONAL FINDINGS

- Jitter: $r = 0.40$ correlation, 1,098 problem cases identified
- Temporal patterns: Only +0.1% improvement (minimal impact)
- RTT and packet loss are independent ($r = 0.105$)
- Statistical significance confirmed ($p < 0.001$ for all tests)

□ RECOMMENDATIONS

- Add TTFB measurements (expected +10-15% R^2)
- Include bandwidth capacity estimates
- Prioritize signal metrics for mobile CDN selection
- For wired networks with RTT+loss only: RTT-only remains optimal